

Roshan Arun Kumar

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Honours (Co-operative Program)

Waterloo, ON

Sept. 2020 – July 2026

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Algorithms, Data Structures and Data Management, Software Design and Architectures, Software and Systems Security, Social Implications of Computing, Operating Systems, Concurrent and Parallel Programming, Computer Networks

EXPERIENCE

Associate Software Developer

AltaML

Sept. 2024 – Dec. 2024

Kitchener, ON

- Owned end-to-end delivery of an autonomous AI agent for the Government of Alberta from ideation through production launch, boosting answer accuracy by 32% over naive prompt baselines by designing an automated document ingestion pipeline with semantic section-aware chunking and hybrid dense-sparse vector retrieval
- Achieved a 98% factual citation rate on high-liability building code inquiries by implementing chain-of-thought grounding prompts, dynamic few-shot domain exemplars, and strict citation-enforcement guardrails
- Cut prompt and retrieval iteration cycles by 65% while preventing production regressions by engineering an automated LLM-as-a-Judge evaluation framework in Python using LangChain, LangSmith, and the OpenAI API across an 80+ scenario test suite calibrated to codes officer standards
- Translated enterprise client requirements into concrete retrieval, chunking, and evaluation criteria by gathering requirements directly from domain experts and communicating design trade-offs to both technical and non-technical stakeholders

Software Engineer

Ford Pro Tech

May 2023 – Aug. 2023

Remote

- Slashed zero-result search queries from 19% to 2.8% and increased search-to-detail click-through rate by 31% across commercial vehicle and charger listings by replacing lexical keyword matching with Elasticsearch dense vector semantic search using transformer text embeddings
- Built and launched the responsive Ford Pro Dealer & Charger Locator across web and mobile, lifting localized dealer lead inquiries by 24% and cutting Largest Contentful Paint by 42% to under 1.6s through geospatial marker clustering, dynamic Mapbox GL integrations, and React/TypeScript performance optimizations

Software Developer

Loblaw Digital

Sept. 2022 – Dec. 2022

Remote

- Engineered a Spring Boot account onboarding microservice for PC Optimum, reducing registration P95 latency by 35% to under 120ms across 12,000+ daily signups by decoupling downstream rewards allocation into asynchronous Kafka events
- Standardized cross-platform REST API contracts and enterprise integrations across 3 client teams (iOS, Android, Web), cutting payload validation errors by 28% by defining OpenAPI 3.0 specifications and implementing centralized Jakarta Bean Validation rules

Software Developer

Home Depot Canada

Jan. 2022 – Apr. 2022

Remote

- Cut daily batch SKU ingestion processing time by 42% while reducing pipeline failure rates from 14% to under 1% by refactoring a synchronous catalog ingest service into an asynchronous Kafka consumer pipeline with exponential backoff retries, Dead Letter Queues, and idempotent database upserts
- Increased automated test coverage from 51% to 87% across 2 core catalog microservices by implementing integration test suites using Testcontainers and WireMock into the CI/CD pipeline, preventing API contract regressions across 4 bi-weekly Agile release cycles

Software Engineering Intern

AdvanceAI

June 2018 – Aug. 2018

Singapore

- Reached 89% recognition accuracy across subject ages on an age-invariant facial recognition task by training a convolutional neural network in Python and TensorFlow

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL (Postgres), NoSQL, C/C++, C#, Swift, Kotlin

AI & Data: LangChain/LangSmith, OpenAI API, RAG, vector databases, LLM evaluation, TensorFlow, scikit-learn, pandas, NumPy, Elasticsearch

Cloud & DevOps: AWS (Lambda, Elastic Beanstalk), Azure, GCP, Docker, CI/CD, REST APIs, microservices, Spring Boot, FastAPI, OpenAPI, Kafka, Redis, React, Git, Linux